

R^2P^3 : Renewal-Reward Process-Based Password Protection in Internet of Things

Thomas Neumann*, Kyle Welsh*, Caden Perez*, Kewei Sha[†], and Yalong Wu*

*Department of Computer Science and Engineering, North Central College, IL, USA

Emails: {tmneumann, kswelsh, cyperez, ywu1601}@noctrl.edu

[†]Department of Computing Sciences, University of Houston–Clear Lake, TX, USA. Email: sha@uhcl.edu

Abstract—Given the eruption of Internet of Things (IoT) systems, it has become critical for connected devices to have persistent security. Password-based security measures provide an interface for users to effectively secure their personal data. Nonetheless, passwords are often stolen by cyber criminals due to inferior user security awareness. To address these issues, in this paper, we develop a renewal-reward process-based password protection (R^2P^3) scheme to prevent unauthorized users from breaching IoT devices even after they obtain user passwords. Specifically, R^2P^3 utilizes the renewal-reward process to accumulate a tendency to trust (T^3) score for any IoT device password attempt on the basis of keystroke dynamics (i.e., dwell time and flight time) and assures device authentication by comparing user attempt T^3 scores to a predefined threshold. We have validated the security efficacy of R^2P^3 . Our experimental results demonstrate that R^2P^3 achieves superior security performance in terms of unauthorized user blocking ratio.

Keywords—Internet of Things, renewal-reward process, password security, tendency to trust score

I. INTRODUCTION

Internet of Things (IoT) systems enable a substantial number of devices (e.g., sensors, actuators, etc.) to communicate through a system of layers (i.e., perception, network, and application) while collecting, measuring, and processing vast amounts of information simultaneously [9], [16]. IoT devices are widely used within most modern technology to provide a convenient platform toward automation. Particularly, IoT systems are becoming widely available to consumers (e.g., Google Nest, Amazon Web Services, etc.) where even consumers' households contain IoT devices. Given the sheer number of these IoT devices, automated security is growing ever more critical, and confidentiality, integrity, privacy, and authentication are all the more necessary [15].

Nonetheless, IoT systems have the realistic possibility of being hindered by unauthorized users due to less attention being placed on them. The number of devices connected to IoT systems could reach seventy-five billion by the year of 2025 [10], and IoT systems will become so substantial that, if maliciously entered, a catastrophic number of devices could be impacted. For example, if the security measures for these IoT devices are weak, unauthorized users could potentially enter billions of security-deficient devices. These instances would impact all markets, from consumer bases to multi-million-dollar corporations. The potential resulting damage is incalculable.

There are a number of research efforts devoted to enhancing the security of IoT systems, and IoT device password protection is one of the top focuses [7], [5], [1], [8], [12], [14], [6]. The existing IoT device password protection research relies on laborious techniques and complex algorithms. Moreover, such techniques perform authentication at levels which are obtainable by more lightweight algorithms. We argue that IoT device password protection should not require complex machine-learning algorithms or neural networks due to the energy constraint of IoT devices. Furthermore, a reasonable amount of these existing works require substantial authorized user input, which burdens users and leads to waning user experience. Thus, it is important to develop techniques to provide IoT systems with a password protection scheme that utilizes a lightweight algorithm and minimal initial authorized user input.

The primary contributions of this work are summarized as follows:

First, we methodically model a defense scheme known as renewal-reward process-based password protection (R^2P^3). The intent of the R^2P^3 scheme is to prevent unauthorized user access to a given IoT system at substantial efficiency. Specifically, R^2P^3 utilizes initial IoT device password input from an authorized user to formulate expected password keystroke timings (i.e., dwell time and flight time). Then, when any individual attempts to access an R^2P^3 protected system, they will be assigned a tendency to trust (T^3) score accumulated by a renewal-reward process, based on how deviated keystroke timings are from the initial input. Thus, if an unauthorized user attempts to access a system they did not setup themselves, they will be denied access because their T^3 score is not above a predefined threshold. It is worth noting that we have designed the R^2P^3 scheme with the concept of it being lightweight in mind. Thus, R^2P^3 was created to perform at high efficiency with minimal data input and only a few statistical calculations.

Second, we devise and perform an experiment that utilizes real participants, each of which took the role of an unauthorized user attempting to access an R^2P^3 protected system. We further prove that the renewal-reward process based password protection scheme is an astonishing solution to the weaknesses of current password security in Internet of Things (IoT) systems. Our experimental results demonstrate that the R^2P^3 scheme is 99.82% effective at denying access to unauthorized users.

The remainder of this paper is organized as follows: In Section II, we introduce network and threat models. In Section III, we present our proposed R^2P^3 scheme in detail. In Section IV, we demonstrate the experimental results that validate the efficacy of the R^2P^3 scheme. In Section V, we discuss the potential enhancements of the R^2P^3 scheme. In Section VI, we review related works. Finally, we conclude the paper in Section VII.

II. NETWORK AND THREAT MODELS

In this section, we present the network model of Internet of Things (IoT) systems and the threat model.

A. Network Model

Internet of Things (IoT) systems typically consist of three layers – perception, network, and application – in which the communication between IoT devices occurs. In the following text, we introduce the three separate layers [3], [9].

Perception Layer. In this layer, IoT devices (e.g., security cameras, temperature gauges, etc.) collect data in a variety of ways based on the intention of the embedded sensors, and, throughout this layer, no data manipulation will take place, as shown in Fig. 1. Specifically, the IoT devices obtain information about some form of action that occurs (e.g., products traveling through an initialized checkpoint). This information may be received either physically from a user or from other devices. Then, it will be sent to and manipulated by the next layer of the IoT system.

Network Layer. In this layer, the collected data from the perception layer is transmitted to a gateway to be processed in a specific manner based on the purpose of the data, as shown in Fig. 1. For example, structured information from radio frequency identification (RFID) tags (attached to IoT objects) could be sent to and received by a separate server in which data is processed and sorted into separate categories.

Application Layer. In this top and final layer of an IoT system, the collected data is stored and analyzed to be converted into meaningful information. The meaningful information now serves a predetermined purpose and can be used within physical objects or applications (e.g., thermostats, smart homes, etc.). As shown in Fig. 1, the application layer is an accessible server, in which we can say data is presented in a formatted manner and IoT devices can freely obtain its structured information.

B. Threat Model

In Section II-A, we identified the perception layer of a typical IoT system. Nonetheless, IoT devices in this layer could be compromised by unauthorized users due to less security and attention. Specifically, certain IoT devices require direct user password input, and unauthorized users may deploy malicious and input-observing software to retrieve excerpts of data (e.g., user credentials) that are of particular importance. This could be achieved by a key-logging program matching the specific machine (i.e., a keyboard) firmware and implemented through the likes of universal serial bus (USB) human interface

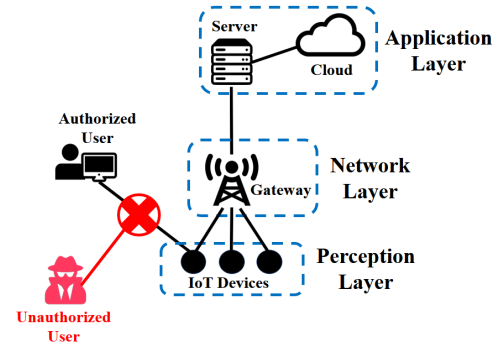


Fig. 1. Network and Threat Models in Internet of Things Systems

devices, which utilize a scripting language in order to modify internal machine settings, manage data pathways, and execute malware payloads [11], [4].

The unauthorized users could inject the keylogging malware into IoT devices through various methods, such as physically inserting an external drive into the device, installation of other malicious programs, or social engineering. After injection, the malware will explicitly search for certain patterns in the user input and thus begin its payload. Specifically, the malware would search for certain keystroke patterns (i.e., a potential password) by the authorized user and store the patterns of interest to a file, database, or server on the system for later use. As shown in Fig. 1, the unauthorized user enters the communication link between the authorized user and an IoT device to observe and record user input.

The unauthorized users could gain a myriad of information on the authorized users, aside from just their passwords. This could include the authorized users' e-mails, phone numbers, and even their social security information in serious cases. This could result in the unauthorized users having direct control over authorized users' banking and financial information, as well as other critical, personal data. Thus, it is imperative to develop a method to prevent these events from occurring, because the outcome of these malware payloads is massive. As such, we endeavor to implement a protection model that prevents the confiscated passwords from being maliciously used by unauthorized users. In the next section, we illustrate our proposed protection scheme against compromised user passwords in Internet of Things systems.

III. R^2P^3 SCHEME

In this section, we present the design rationale and the key components of our scheme.

A. Design Rationale

Our proposed renewal-reward process-based password protection (R^2P^3) scheme endeavors to block unauthorized users from breaching Internet of Things (IoT) devices even after they have compromised the password of the IoT device. Particularly, R^2P^3 utilizes keystroke dynamics dwell time (i.e., the duration that a key is pressed) and flight time (i.e., the duration between keystrokes) to introduce unique biometric

features to the password of the IoT device and thus strengthen its security against cyber threats.

Specifically, R^2P^3 considers two biometrics (i.e., dwell and flight times) in each password keystroke. R^2P^3 will use the renewal-reward process to accumulate a tendency to trust (T^3) score for any user's (authorized or unauthorized) password attempt, and it will assure the IoT device will only authenticate the authorized user. With the integration of biometrics, R^2P^3 ensures the integrity and confidentiality of IoT systems.

Our proposed scheme consists of two key components: (i) **Password setup** constructs biometric information for the standard password of an IoT device with the authorized user password attempts during account setup, and (ii) **Password protection** utilizes the renewal-reward process to compute a T^3 score for each IoT device password attempt on the basis of the biometric information, and protects IoT device from threats and breaches. In the following sections, we describe these components in detail.

B. Password Setup

It is exceptionally important for the authorized users of Internet of Things devices to set up a strong password to protect their personal data against cyber criminals. R^2P^3 requires authorized users to enter the password of choice multiple times during the account setup. This assists R^2P^3 to better assemble the user's biometric information, in our case dwell and flight times, of the authorized user, and include that information in each keystroke of the password.

With the illustration in Section III-A, we understand that there are three components in each keystroke of the password: a character on the keyboard, the dwell time on the character, and the flight time from the release of this character to the press of the next one. Assume that the number of keystrokes in the password of choice from the authorized user is N_κ , then it shall be bounded between a minimum positive integer $N_{\min}$ (e.g., 8) and a maximum positive integer $N_{\max}$ (e.g., 14). It is worth noting that both $N_{\min}$ and $N_{\max}$ shall be decided by factory settings of the IoT device. If we denote a keystroke in the password as $\mathbf{K}_i$ ($0 \leq i \leq N_\kappa - 1$), then $\mathbf{K}_i$ will be a row vector with three columns and could be represented as $\mathbf{K}_i = (C_i, T_{d_i}, T_{f_i})$, where C_i is the i^{th} character in the password, T_{d_i} is the dwell time of C_i , and T_{f_i} is the flight time of C_i , respectively.

During the account registration, if R^2P^3 demands an authorized user to enter the password of choice N_ϵ ($N_\epsilon = 1, 2, 3, \dots$ and $N_\epsilon = 1$ shall be avoided) times, then each entry of the password can be denoted as $\mathbf{E}_j$ ($0 \leq j \leq N_\epsilon - 1$). $\mathbf{E}_j$ is also a row vector with $3N_\kappa$ columns and can be represented as follows:

$$\mathbf{E}_j = (\mathbf{K}_0^j, \mathbf{K}_1^j, \dots, \mathbf{K}_{N_\kappa-2}^j, \mathbf{K}_{N_\kappa-1}^j), \quad (1)$$

where $\mathbf{K}_i^j = \{C_i, T_{d_i}^j, T_{f_i}^j\}$ since each password entry could be slightly different on dwell time and flight time. If we include all the password entries from the authorized user together, then we will have a password matrix as follows:

$$\mathbf{PWS} = (\mathbf{E}_0^T, \mathbf{E}_1^T, \dots, \mathbf{E}_{N_\epsilon-2}^T, \mathbf{E}_{N_\epsilon-1}^T)^T, \quad (2)$$

there are N_ϵ rows and $3N_\kappa$ columns in **PWS**.

With the password matrix **PWS**, it is critical to derive a standard password of biometric information for the IoT device to validate any access attempts. It is well known that arithmetic mean is the central value of a series of numbers, and standard deviation is the amount of variations of a series of numbers. They capture and represent the key characteristics of a data set. Thus, our proposed R^2P^3 scheme continues to compute a biometric vector of the means and standard deviations of the dwell time and flight time for each keystroke in the password and uses it to assess the biometric information from any access attempts to enforce IoT device authentication.

Specifically, R^2P^3 averages the N_ϵ number of dwell times $T_{d_i}^j$ ($j = 0, 1, \dots, N_\epsilon - 1$) from **PWS** to get their mean value μ_{d_i} , which will be used as the standard dwell time for character C_i in the standardized password. In addition, R^2P^3 also utilizes μ_{d_i} to obtain a standard deviation σ_{d_i} of dwell times for each character C_i to support extra password validation. The derivation of both μ_{d_i} and σ_{d_i} is as follows:

$$\mu_{d_i} = \frac{1}{N_\epsilon} \sum_{j=0}^{N_\epsilon-1} T_{d_i}^j, \quad \sigma_{d_i} = \sqrt{\frac{\sum_{j=0}^{N_\epsilon-1} (T_{d_i}^j - \mu_{d_i})^2}{N_\epsilon}}. \quad (3)$$

Likewise, the standard flight time μ_{f_i} in the standard password and the corresponding standard deviation σ_{f_i} for character C_i can be derived from the flight times $T_{f_i}^j$ ($j = 0, 1, \dots, N_\epsilon - 1$) of **PWS** and represented as follows:

$$\mu_{f_i} = \frac{1}{N_\epsilon} \sum_{j=0}^{N_\epsilon-1} T_{f_i}^j, \quad \sigma_{f_i} = \sqrt{\frac{\sum_{j=0}^{N_\epsilon-1} (T_{f_i}^j - \mu_{f_i})^2}{N_\epsilon}}. \quad (4)$$

If we denote the biometric vector for each keystroke $\mathbf{K}_i$ in the standard password of the authorized user as $\mathbf{B}_i$, then $\mathbf{B}_i = (\mu_{d_i}, \sigma_{d_i}, \mu_{f_i}, \sigma_{f_i})$. $\mathbf{B}_i$ will be utilized in the next section to measure the distance between the standardized biometric information (i.e., T_{d_i} and T_{f_i}) of $\mathbf{K}_i$ and the biometric information from any IoT device password attempt. The device password attempt is only deemed to be from the authorized user when the distance falls under a R^2P^3 defined threshold. Otherwise, the IoT device password attempt will be denied.

C. Password Protection

In this work, we assume that the sequence of characters S_c in the password of the authorized user has been compromised and the unauthorized user has obtained $S_c = (C_0, C_1, \dots, C_{N_\kappa-2}, C_{N_\kappa-1})$. Thus, it is critical for our proposed R^2P^3 scheme to utilize the biometric information of dwell time and flight time to distinguish the authorized user from the unauthorized user, which will be blocked from accessing to the IoT device.

Specifically, R^2P^3 first introduces Z-score to measure the distance between the standardized biometric information (i.e., dwell time mean μ_{d_i} and flight time mean μ_{f_i}) of $\mathbf{K}_i$ from password setup by the authorized user and that (i.e., $\widehat{T}_{d_i}$ and

$\widehat{T}_{f_i}$) of the i^{th} keystroke $\widehat{\mathbf{K}}_i$ ($\widehat{\mathbf{K}}_i = (C_i, \widehat{T}_{d_i}, \widehat{T}_{f_i})$) of any IoT device password attempt. If we denote the Z -scores (i.e., distance) between μ_{d_i} and $\widehat{T}_{d_i}$, between μ_{f_i} and $\widehat{T}_{f_i}$ as $\widehat{Z}_{d_i}$ and $\widehat{Z}_{f_i}$, then they can be derived by using the biometric vector $\mathbf{B}_i = (\mu_{d_i}, \sigma_{d_i}, \mu_{f_i}, \sigma_{f_i})$ of $\mathbf{K}_i$ from Section III-B, and the definition of Z -score is as follows:

$$\widehat{Z}_{d_i} = \frac{\widehat{T}_{d_i} - \mu_{d_i}}{\sigma_{d_i}}, \quad \widehat{Z}_{f_i} = \frac{\widehat{T}_{f_i} - \mu_{f_i}}{\sigma_{f_i}}, \quad (5)$$

where $\widehat{Z}_{d_i} = 0$ indicates that the Z -score of $\widehat{T}_{d_i}$ is the same as that of μ_{d_i} and $\widehat{Z}_{d_i} = 1$ infers that $\widehat{T}_{d_i}$ is one standard deviation away from μ_{d_i} . In addition, $\widehat{Z}_{d_i}$ could be positive or negative. A positive $\widehat{Z}_{d_i}$ implies that $\widehat{T}_{d_i}$ is above the mean μ_{d_i} , while a negative value tells us that $\widehat{T}_{d_i}$ is below the mean. The same logic also applies to $\widehat{T}_{f_i}$. Thus, we could have a very good sense of how distant the biometric information is from that of the standardized IoT device password.

In our proposed R^2P^3 scheme, the dwell time T_{d_i} follows a normal distribution with mean value μ_{d_i} and standard deviation σ_{d_i} , which will be proved shortly through experimentation in Section IV. Likewise, the flight time T_{f_i} also follows normal distribution with mean value μ_{f_i} and standard deviation σ_{f_i} . It is worth noting that we have made a weak assumption about the distributions of $\widehat{T}_{d_i}$ and $\widehat{T}_{f_i}$. They are both assumed to come from the same probability distributions (i.e., normal distribution) as T_{d_i} and T_{f_i} . Thus, the Z -scores (i.e., $\widehat{Z}_{d_i}$ and $\widehat{Z}_{f_i}$) in Equation (5) are both standardized normal distributions with means of 0 and standard deviations of 1. Please refer to Section IV-A for a detailed explanation.

In this paper, we interpret Z -score as a metric of tendency to trust (T^3) score. Specifically, if the biometric information (i.e., $\widehat{T}_{d_i}$ and $\widehat{T}_{f_i}$) from any device password attempt is close to the biometric information mean values (i.e., μ_{d_i} and μ_{f_i}) from the authorized user, then the T^3 scores (i.e., $\widehat{Z}_{d_i}$ and $\widehat{Z}_{f_i}$) will be near 0 and the password attempt will be more likely to be trusted. Otherwise, it will be further away from 0 and the password attempt will be unlikely to be trusted. In order to adjust the sensitivity level of the T^3 score, a Z -score threshold could be created. To be specific, the Z -score has to be a greater magnitude than the sensitivity level for the T^3 score to receive a positive value and a lesser magnitude for the T^3 score to receive a negative value. Please refer to Equation (5) for details. This allows us to better understand how deviated $\widehat{Z}_{d_i}$ and $\widehat{Z}_{f_i}$ really are from the μ_{d_i} and μ_{f_i} , respectively. If we denote the Z -score threshold as Ψ , then the T^3 score for each $\widehat{T}_{d_i}$ and $\widehat{T}_{f_i}$ could be derived as follows:

$$\widehat{T}_{d_i}^3 = \widehat{Z}_{d_i} - \Psi, \quad \widehat{T}_{f_i}^3 = \widehat{Z}_{f_i} - \Psi. \quad (6)$$

It is worth noting that $\widehat{T}_{d_i}^3$ and $\widehat{T}_{f_i}^3$ both follow normal distribution with the mean of $-\Psi$ and standard deviation of 1, because only constant Ψ is subtracted from $\widehat{Z}_{d_i}$ and $\widehat{Z}_{f_i}$, and they both abide by standard normal distribution. Thus, $\widehat{T}_{d_i}^3$ and $\widehat{T}_{f_i}^3$, $i = 0, 1, \dots, N_\kappa - 1$, are independent and identically (*i.i.d*) distributed variables.

With the description in Section III-B, we understand that there are N_κ keystrokes in the password of the authorized IoT device user. Thus, it is a necessity to accumulate the T^3 scores of all the keystrokes from the IoT device password attempt. The overall tendency to trust score helps us to decide if the IoT device access attempt will be granted. In our proposed R^2P^3 scheme, we utilize the renewal-reward process [13] to assemble the tendency to trust score, because it is a well-behaved probability distribution and exists in most physical objects. Specifically, the renewal-reward process computes an accumulated *i.i.d* random reward (i.e., $\widehat{T}_{d_i}^3$ score and $\widehat{T}_{f_i}^3$ score) at the end of each keystroke on the basis of *i.i.d* inter renewal times ($\widehat{T}_{d_i}$ and $\widehat{T}_{f_i}$). Thus, the renewal-reward process includes accumulated tendency to trust by the time of each keystroke. It is worth noting that we have assumed that the dwell time and flight time (as the inter renewal times) are independently and identically distributed variables. If we denote the overall tendency to trust score accumulated by the end of any IoT device password attempt as $\widehat{T}^3$, then it could be represented as follows:

$$\widehat{T}^3 = \sum_{i=0}^{N_\kappa-1} (\widehat{T}_{d_i}^3 + \widehat{T}_{f_i}^3). \quad (7)$$

We will then compare $\widehat{T}^3$ with a tendency to trust threshold Φ (say $\widehat{T}^3 \geq \Phi$, this shall be discussed and set shortly in Section IV), and the result from that comparison will determine if the password attempt will be granted access to the IoT device. The threshold Φ could also be multiplied by N_κ to scale the threshold with the length of the password. For example, $\widehat{T}^3 \geq N_\kappa \Phi$. This is beneficial since a longer password has more opportunities for a positive tendency to trust score. It is worth noting that R^2P^3 is very lightweight, because the calculations in R^2P^3 only consist of summations and standard deviations, whose time complexity is not significant.

IV. PERFORMANCE EVALUATION

In this section, we demonstrate the experimental results of our proposed renewal-reward process-based password protection (R^2P^3) scheme with regard to unauthorized user blocking ratio in Internet of Things (IoT) systems, which is directly related to the tendency to trust (T^3) score. The R^2P^3 scheme in this paper is implemented in JavaScript.

A. Experimentation Setup

Two laptops were used during the testing of R^2P^3 . The first laptop is a Lenovo Yoga 6 (13.3"), which contains an AMD Ryzen 7 5700U processor. The second laptop is a MSI Summit E13 FLIP (13.4"), which contains an Intel Core i7-1185G7 processor. It is of great importance to note that although two separate laptops were used in our experiment, they contained the same keyboard style: 75%. All keys utilized while participants would type a given password would have the same relative position on both machines.

The initial password matrix **PWS** was created by observers through the entry of a specific password ten times through the **Password setup** process examined earlier in Section III-B.

TABLE I
PASSWORDS

Length	Password
8	Movies1*
9	Videos14@
10	Password9=
10	Timezone1#
11	Abcdefg712
12	Password123
13	Computeryou8?
14	Itsaniceday68\$

The entry amount of ten times was chosen because it would not be reasonable to ask authorized users to perform an extended password setup process through a large series of entries (say 20) in real life. The password length in our experimentation was set between 8 and 16, because of the nature of current password requirements throughout the real-world systems (e.g., email and banking systems). Each password would contain a series of characters, a few character numbers (e.g., 1, 2, etc.), and one symbol (e.g., *, !, etc.). Thus, it was predetermined that the passwords denoted in Table I would be utilized in the experimentation.

B. Data Preparation

1) *Data Rationality*: The duration in which a user holds down a key (i.e., dwell time) and the duration between each user keystroke (i.e., flight time) would be utilized as the input of keystroke dynamics for R^2P^3 , because the typing style of each individual is relatively unique. Specifically, each individual possesses different hand and finger sizes which will have an impact on typing style and timings. Therefore, dwell and flight times can be attempted to be utilized to differentiate between system users.

2) *Data Collection*: Data collection took place at North Central College (NCC, Naperville, IL, USA) from December, 2021 to January, 2022. Specifically, we have collected the IoT device password attempt data from NCC students and professors, which were known as unauthorized users throughout our experiment. In total, 95 unauthorized users were given the opportunity to attempt to access the R^2P^3 protected IoT system. Testing could possibly be done on more individuals in the future. However, 95 individuals were all we could obtain for this series of testing.

In data collection, an unauthorized user would be presented with the task of infiltrating the R^2P^3 protected IoT system. Specifically, the unauthorized users would be made knowledgeable that they have already obtained the password of the R^2P^3 protected system. Then, each of the unauthorized users was provided with the sequence of characters, numbers, and symbols that make up a specific password, and was required to attempt to access the R^2P^3 protected IoT system a certain number of times (e.g., 5, 6, 7, etc.).

C. R^2P^3 Evaluation

With the illustration in Section III-C, we understand that the input data (i.e., dwell and flight times) of R^2P^3 must

follow normal distribution. In our experiment, we have utilized Anderson-Darling test [2] to validate the normality of the keystroke dynamics. Specifically, p-values were calculated after data preprocessing in which the well know IQR (interquartile range) statistical method was used on data sets of various password attempts by an authorized user from Section III-B. A p-value greater than or equal to 0.05 implies that there is no significant difference between a specific keystroke data set and a typical normal distribution, while any p-value less than 0.05 implies a significant difference. As shown in Fig. 2, most of the p-values are above the baseline of 0.05, and there are only a few p-values under the baseline. Thus, we can make the conclusion that the dwell and flight times follow normal distribution. It is important to note that Fig. 2 displays the p-values of all sets of dwell and flight times of the passwords in Table I.

All outcomes in our experiment were obtained while Z-score threshold $\Psi = 1.2$. This was predefined through initial data collection to calculate the optimal Ψ . In this process, a specific user was required to set up an account and Ψ was then modified in the range of (0.1, 0.2, ..., 2.9). Each time Ψ was modified, a consistent unauthorized user attempted to breach the R^2P^3 protected IoT system multiple times (e.g., 3). Then, the average T^3 score of the unauthorized user for a specific Ψ value was recorded. Fig. 3 demonstrates the corresponding data for this specific test. As we can see, 1.2 is when the average score of a specific authorized user remains above the T^3 (tendency to trust) score threshold Φ , and this is also consistent with the average score of the unauthorized user remaining below Φ . Thus, we can examine that the optimal value for Ψ is 1.2.

In our experiment, the performance of R^2P^3 is evaluated in terms of tendency to trust (T^3) score with regard to number of attempts, keystroke amount, and password length.

Fig. 4 demonstrates the average T^3 score of the unauthorized user over diverse number of IoT device password attempts. As we can see, the average T^3 score of unauthorized user increases when more password attempts are given to breach the IoT system. Nonetheless, it is worth noting that all T^3 scores are still under the threshold $\Phi = 0$. Thus, the unauthorized users are not able to compromise R^2P^3 protected IoT system even after a vast number of password attempts are allowed.

Fig. 5 shows that the T^3 score of unauthorized user decreases as the amount of keystrokes in a specific password increases. This makes an R^2P^3 protected IoT system even more difficult to breach when there are more keystrokes in the password. Thus, the amount of keystrokes in a password has a significant impact on access allowance of an IoT device. It is worth noting that R^2P^3 also detects which keystroke was used (e.g., left shift or right shift). Specifically, if the IoT device password of an authorized user contains a left shift, and an unauthorized user replaces that specific left shift with a right shift, the unauthorized user will be denied access and assigned no T^3 score. Thus, unauthorized users will have no knowledge if they are even attempting to breach the IoT system with the

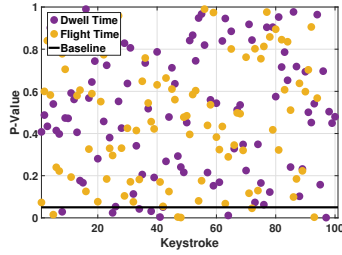


Fig. 2. Keystroke Dynamics Fitness Validation

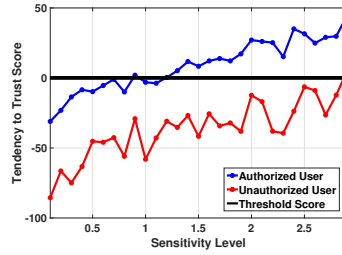


Fig. 3. Optimal Z-Score Threshold Selection

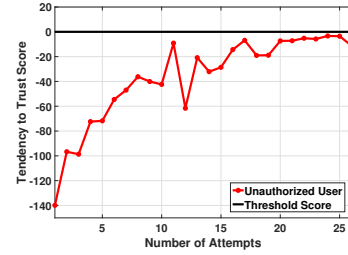


Fig. 4. T^3 Score Across Attempts

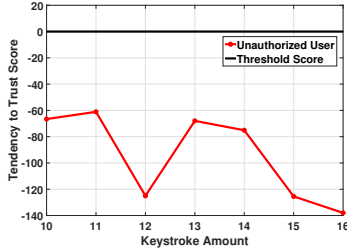


Fig. 5. T^3 Score Across Keystroke Amount

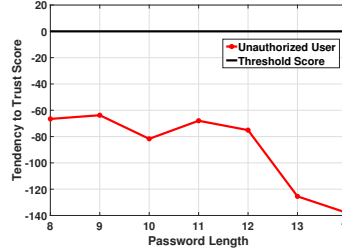


Fig. 6. T^3 Score Across Password Length

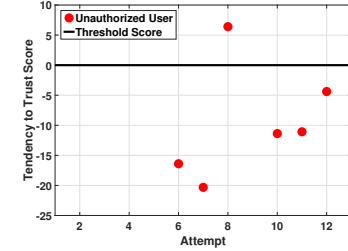


Fig. 7. Breaching User T^3 Scores

correct keys. In our experiment, 56.97% of unauthorized users were blocked in this manner.

Fig. 6 presents experimental result of T^3 scores comparing the amount of characters that make up a password, ignoring keys not physically present in the password (e.g., left shift and right shift). As we can see, the value of T^3 score decreases significantly when the password length increases. Thus, IoT system security is increased drastically with our proposed R^2P^3 scheme.

In our experiment, there is indeed an attempt that breached the R^2P^3 protected IoT system, and Fig. 7 presents the said attempt. The lack of data on a few attempts in Fig. 7 is because of the user using an incorrect key not present in the password (e.g., left shift, right shift, etc.). This means the user was assigned no T^3 score, as explained previously. As shown in the figure, the attempts by this specific unauthorized user still remain scattered. Thus, the unauthorized user could not gain access to the system consistently even after the R^2P^3 model is breached. This further proves the inability of unauthorized users to consistently gain access while the R^2P^3 model is present in an IoT system. It is worth noting that this successful unauthorized user access was performed while the password length was 8. Thus, the only attempt in which an unauthorized user accessed the R^2P^3 protected IoT system was an instance where the password length met the bare minimum requirements. Nonetheless, no unauthorized users in our experiment were able to access the R^2P^3 protected IoT system, while the password length is greater than 8.

In our experiment, the overall unauthorized user blocking ratio is 99.82%. Specifically, our renewal-reward process-based password protection (R^2P^3) scheme has blocked 575 unauthorized access attempts out of 576 unauthorized access attempts. Thus, R^2P^3 is an obvious concrete solution for device password protection in Internet of Things systems.

V. DISCUSSION

In this section, we discuss several strategies to enhance our proposed renewal-reward process-based password protection (R^2P^3) scheme in Internet of Things (IoT) systems. The experimental data yields great results for the R^2P^3 scheme. Nonetheless, there are still methods that the unauthorized users could employ to bypass this layer of protection. One particularly realistic issue is that the unauthorized user could collect the IoT device password dwell and flight timings of the authorized user, and simulate these timings when entering credentials. This would make the device authentication trivial. Throughout the following text, we describe several refinements that could be implemented into our R^2P^3 scheme in order to add more layers of protection to this caveat.

Dynamic Timing Update. In this refinement, the timings (i.e., dwell and flight times) of the authorized users passwords shall be dynamically updated in real time. Specifically, an average dwell time and an average flight time for each keystroke by the authorized user should be calculated each time they log in to their respective IoT device account. Over time as the authorized user enters the password to an IoT system, these averages will update to match both their historical records as well as their current records. Consequently, the stored keystroke information of the authorized user will constantly be updated and the unauthorized users will thus possess outdated information.

Device Account Lockup. In this refinement, R^2P^3 would include another algorithm that activates when suspicious activity from an unauthorized user is detected. There are several implementations of this kind, such as locking the IoT device account if the keystroke timings of the unauthorized user are too closely aligned to those of the authorized user. The account could also lock by detecting the static information (e.g., internet-protocol address, type of device, etc.) from the unauthorized users when they are attempting to log in to the IoT system. Lastly, the account could lock after a flat

integer amount of login attempts are exceeded. Specifically, the unauthorized user must have a high enough tendency to trust score to successfully log in to the account before the threshold entries are exceeded. To sum up, if any of these scenarios occur, the unauthorized user will no longer be prompted to enter any credentials to the IoT system.

Random Password Pattern. In this refinement, a certain number of keystroke patterns of the authorized user would be stored in the IoT system. This would occur during the password setup stage of the R^2P^3 scheme. Each time an IoT device attempt is made to authenticate, one of these patterns will be used to compare with the given attempt. These patterns would be chosen from based on a random integer that is generated upon each submitted attempt in order to find one of the several patterns. If an unauthorized user has one data set of keystroke timings, it will not be consistent enough to compare with all of the diverse stored patterns of the authorized user. Thus, the unauthorized user will not be able to consistently authenticate with the keystroke timings they possess.

VI. RELATED WORKS

A number of research efforts have been devoted to password protection on the basis of keystroke dynamics in Internet of Things systems [7], [5], [1], [8], [12], [14], [6]. These existing works can be classified into the category of initial authentication schemes, in which only an one-time comparison is performed against any device password attempt. For example, Huang *et al.* [7] utilized three machine learning classifiers (i.e., support vector machine, artificial neural network, and random forest) to analyze collected user touch times and force features to harden device passwords in Internet of Things systems. This scheme performed with an equal error rate of 0.720%. Bartlow *et al.* [1] formulated a scheme in which the random forest technique was employed in combination with keystroke dynamics (e.g., key latency) to provide username and password authentication. The scheme achieved a 1% false acceptance rate while attaining a 14% false reject rate.

Unlike the existing schemes, our proposed renewal-reward process-based password protection (R^2P^3) scheme utilizes renewal theory in statistics. Thus, it provides new evidence towards the efficiency of different models for device password protection in Internet of Things systems. R^2P^3 is lightweight compared to more complex models for password protection. Our work also consists of thorough real-life based testing without the use of simulation.

VII. CONCLUSION

In this paper, we developed a renewal-reward process-based password protection (R^2P^3) scheme to prevent unauthorized users from breaching Internet of Things (IoT) devices even after their passwords are compromised. Specifically, R^2P^3 utilizes renewal-reward process in combination with keystroke dynamics (i.e., dwell and flight times) to assemble an overall tendency to trust (T^3) score for each IoT device password attempt, and the attempt is only granted access to IoT system

when its T^3 score exceeds a predefined threshold. Our experimental results demonstrated that R^2P^3 is an obvious solution to IoT password protection in terms of the unauthorized user blocking ratio.

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